

Workshop

Ekarat Rattagan
NIDA

Outline

1. Databricks
2. Google Bigquery

DATA + AI SUMMIT HAPPENING NOW

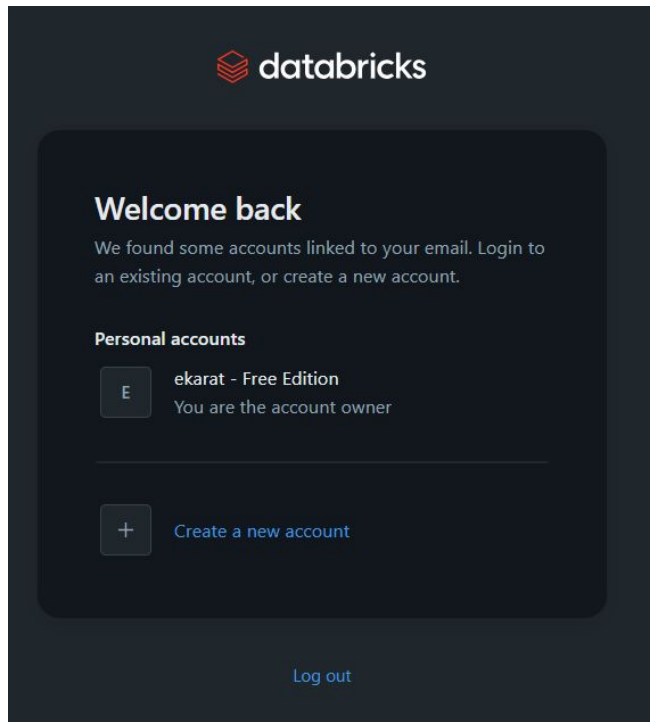
Don't miss the free livestream to hear from leading experts, researchers and open source contributors.

[Register now](#)The logo for the Databricks DATA+AI SUMMIT. It features the Databricks logo (a stylized cube) followed by the word "databricks" in a lowercase sans-serif font. Below this, the words "DATA+AI" and "SUMMIT" are stacked in a large, bold, uppercase sans-serif font. The entire logo is set against a dark red background with a large, stylized hexagonal shape behind it.

databricks
**DATA+AI
SUMMIT**

JUNE 15-18 | SAN FRANCISCO

Setup



+ New

 Home

 Learn

 Workspace

 Recents

 Catalog

 Jobs & Pipelines

 Compute

 Discover Beta

 Marketplace

SQL

 SQL Editor

 Queries

 Dashboards

 Genie Spaces

 Alerts

 Query History

 SQL Warehouses

Data Engineering

 Runs

Welcome to Databricks

 Send feedback

Day streak
 2

Courses taken
 0

Data sets created
 0

Friends added
 0

Start learning >

Courses

 **Databricks Fundamentals**
Beginner

 **Get Started with Data Engineering**
Beginner

 **Intro to AI/BI**
Intermediate

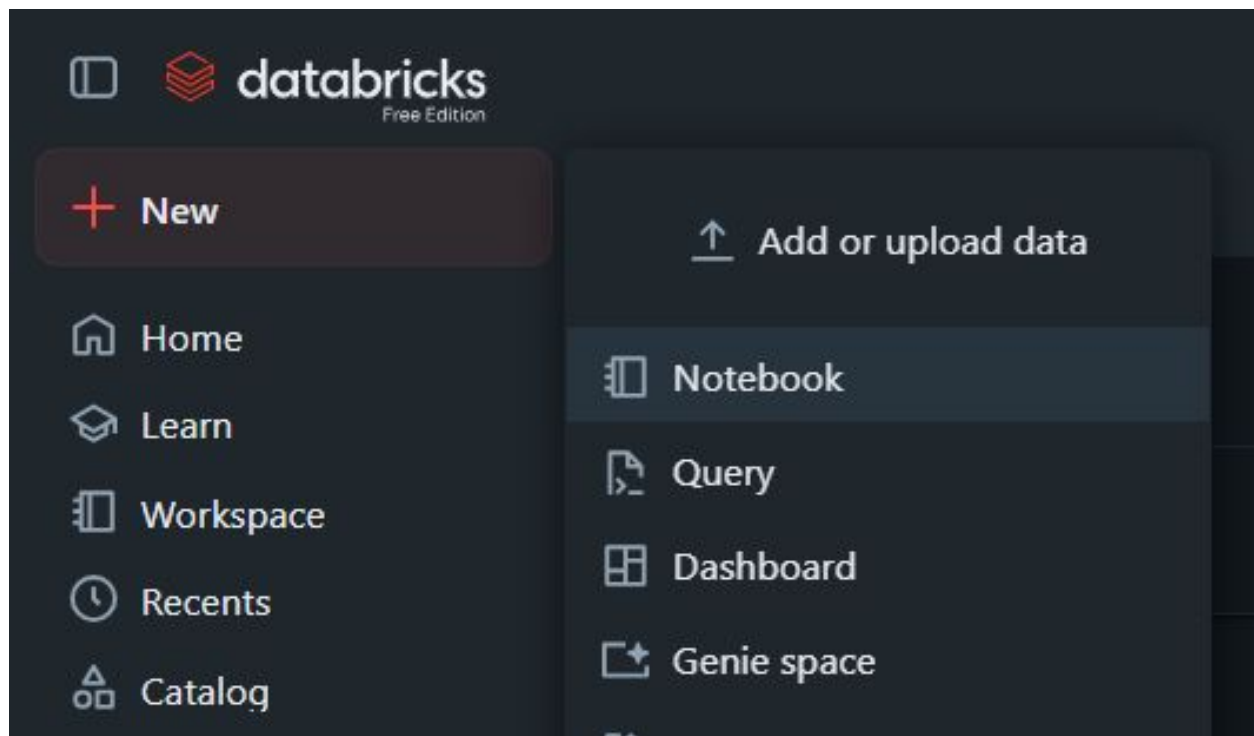
Interactive demos

 **Clean data with a pipeline**
Build an ETL pipeline to clean bike share data

 **CDC Pipeline With Delta**
CDC flow with Spark API and Delta Lake

 **Visualize data in a dashboard**
Build a dashboard from bike share data

Your resources



The screenshot displays the Databricks Free Edition web interface. On the left is a dark sidebar with navigation links: Home, Learn, Workspace (highlighted), Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), and Marketplace. Below these are links for SQL, SQL Editor, Queries, and Dashboards. The top header features the Databricks logo and a search bar labeled "Search data, notebooks, recents, and more...". The main workspace area is divided into three panels. The left panel shows a file tree under the "Workspace" tab, with "Tree view: ON" and a breadcrumb path of ".../ Drafts". It lists a "New Notebook 2026-06-17 14:..." (selected), "note1", and "sample_data.csv". The middle panel shows a menu with "File", "Edit", "View", "Run", "Help", a "Python" language selector, a star icon, and a link "Last edit was now". The right panel is the notebook editor, featuring a blue play button icon and the text "Start typing or generate with AI (Ctrl + I)...". At the bottom right, a banner reads "Let Genie Code turn thoughts into results".

Experiment

1. Dataset

https://github.com/ekaratnida/Data-Engineer/blob/main/bdi-mu/sample_data.csv

2. Notebook (Pyspark):

<https://github.com/ekaratnida/Data-Engineer/blob/main/bdi-mu/note1.ipynb>

3. RDD (Use colab)

https://github.com/ekaratnida/Data-Engineer/blob/main/de/bdi-mu/RDD_Version.ipynb

2. BigQuery

The screenshot shows the Google Cloud BigQuery documentation page for the sandbox. The page has a light blue header with the Google Cloud logo, navigation links for Technology areas, Cross-product tools, and a search bar. Below the header is a sub-header with the BigQuery logo and navigation links for Overview, Guides (selected), Reference, Samples, and Resources. A left sidebar contains a 'Filter' button and a list of categories: Discover (Product overview, Try BigQuery using the sandbox), Get started (Console walkthroughs and videos, Use the console, Use the bq CLI tool, Use the client libraries), Plan (Organize resources, API dependencies, Datasets, Tables, Views), Load, transform, and export (Introduction, Build a data pipeline with Data Engineering Agent, Migrate data, Load data, Transform data, Export data, ELT tutorials), and Analyze. The main content area shows the breadcrumb 'Home > Documentation > Data analytics > BigQuery > Guides', a 'Was this helpful?' feedback section, and the title 'Try BigQuery using the sandbox'. The text explains that the BigQuery sandbox allows users to explore limited capabilities at no cost. It also mentions that Google Cloud offers public datasets available through the Google Cloud Public Dataset Program. A 'Guide me' button is provided for step-by-step guidance. The section 'Before you begin' includes the instruction to 'Enable the BigQuery sandbox' and a numbered list starting with '1. In the Google Cloud console, go to the BigQuery page.' followed by a 'Go to BigQuery' button. At the bottom of this section, it states that users can also open BigQuery in the Google Cloud console by entering a specific URL in their browser.

Google Cloud Technology areas Cross-product tools Search

BigQuery Overview Guides Reference Samples Resources

Filter

Discover
Product overview
Try BigQuery using the sandbox

Get started
Console walkthroughs and videos
Use the console
Use the bq CLI tool
Use the client libraries

Plan
Organize resources
API dependencies
Datasets
Tables
Views

Load, transform, and export
Introduction
Build a data pipeline with Data Engineering Agent
Migrate data
Load data
Transform data
Export data
ELT tutorials

Analyze

Home > Documentation > Data analytics > BigQuery > Guides Was this helpful? Send feedback

Try BigQuery using the sandbox

The BigQuery sandbox lets you explore limited BigQuery capabilities at no cost to confirm whether BigQuery fits your needs. The BigQuery sandbox lets you experience BigQuery without providing a credit card or creating a billing account for your project. If you already created a billing account, you can still use BigQuery at no cost in the free usage tier.

The BigQuery sandbox lets you learn BigQuery with a limited set of BigQuery features at no charge. You can evaluate BigQuery by using the BigQuery sandbox to view and query a public dataset.

Google Cloud offers public datasets that are stored in BigQuery and made available to the general public through the [Google Cloud Public Dataset Program](#). For more information about working with public datasets, see [BigQuery public datasets](#).

To follow step-by-step guidance for this task directly in the Google Cloud console, click **Guide me**:

Guide me

Before you begin

Enable the BigQuery sandbox

1. In the Google Cloud console, go to the **BigQuery** page.

Go to BigQuery

You can also open BigQuery in the Google Cloud console by entering the following URL in your browser:

<https://docs.cloud.google.com/bigquery/docs/sandbox>

BigQuery: play and learn

The screenshot displays the Google Cloud BigQuery Studio interface. At the top, the Google Cloud logo and 'My Project' are visible. A search bar is located on the right. The left sidebar, titled 'Explorer', contains a search bar and a list of navigation items: Home, Starred, Shared with me, Job history, and a project named 'rich-atom-320803'. Under this project, there are links to Datasets, Connections, Conversations, Agents, and (Classic) Queries. The main workspace features a blue header with the text 'Check out what's new in Studio'. Below this, a list of features is shown: 'Chat with data' (highlighted), Bulk actions in Files, Cloud Assist, Data preparation, and Query parameters. To the right of this list is a preview of the 'Chat with data' feature, showing a chat interface with a 'Chat with table' button and a line chart. Further right, text describes the feature: 'Get insights by chatting with your data' (with a 'Preview' button), 'Chat with your data to generate, execute, and visualize answers grounded in questions or run the generated code to verify results.', and 'Set up a new agent by connecting it to datasets or start a new chat directly'. A bullet point states: 'To activate this feature, you need to enable the Dataform API.' Below this text are 'Enable APIs' and 'Next' buttons. At the bottom, a 'Create new' bar offers options: 'SQL query', 'Notebook', 'Notebook with Spark' (with a dropdown arrow), 'ML model', 'Graph', 'Agent', 'Conversation', 'Pipeline', and 'Data'.

Google Cloud My Project

Search (/) for resources, docs, products, and more Search

Explorer + Add data

Search for resources

Home

Starred

Shared with me

Job history

rich-atom-320803

Datasets

Connections

Conversations

Agents

(Classic) Queries

Check out what's new in Studio

Chat with data

Bulk actions in Files

Cloud Assist

Data preparation

Query parameters

Chat with table

Get insights by chatting with your data Preview

Chat with your data to generate, execute, and visualize answers grounded in questions or run the generated code to verify results.

Set up a new agent by connecting it to datasets or start a new chat directly

- To activate this feature, you need to enable the Dataform API.

Enable APIs Next

Create new

SQL query Notebook Notebook with Spark ML model Graph Agent Conversation Pipeline Data

BigQuery: play and learn

The screenshot shows the Google Cloud BigQuery Studio interface. At the top, there's a Google Cloud logo and a 'My Project' dropdown. A search bar is on the right. Below the top bar is a navigation bar with icons for Explorer, Recent, and other views. The Explorer sidebar on the left contains a search bar and a list of resources: Home, Starred, Shared with me, Job history, and a project named 'rich-atom-320803'. Under this project, there are links for Datasets, Connections, Conversations, Agents, and (Classic) Queries. A red arrow with the number '1' points to the '+ Add data' button in the Explorer sidebar. The main content area displays a 'Check out what's new in Studio' section with a list of features: Chat with data, Bulk actions in Files, Cloud Assist, Data preparation, and Query parameters. A preview window shows a 'Chat with table' interface with a line chart. To the right of the preview, there's text about getting insights by chatting with data, a 'Preview' button, and instructions on how to set up a new agent. At the bottom, there's a 'Create new' section with buttons for SQL query, Notebook, Notebook with Spark, ML model, Graph, Agent, Conversation, Pipeline, and Data.

Google Cloud My Project

Search (/) for resources, docs, products, and more Search

Explorer + Add data

Search for resources

Home

Starred

Shared with me

Job history

rich-atom-320803

Datasets

Connections

Conversations

Agents

(Classic) Queries

Check out what's new in Studio

Chat with data

Bulk actions in Files

Cloud Assist

Data preparation

Query parameters

Get insights by chatting with your data Preview

Chat with your data to generate, execute, and visualize answers grounded in questions or run the generated code to verify results.

Set up a new agent by connecting it to datasets or start a new chat directly

- To activate this feature, you need to enable the Dataform API.

Enable APIs Next

Create new

SQL query Notebook Notebook with Spark ML model Graph Agent Conversation Pipeline Data

Google Cloud My First Project

Add data Data sources

Filter By

Data Source Type

- ☐ Databases (19)
- ☐ Marketing (7)
- ☐ Business Applications (26)
- ☐ Storage/Data Lakes (6)
- ☐ Data Warehouses (3)
- ☐ Data Streaming (4)

Public Datasets 2

Sharing (Analytics Hub)

Star a project by name

Product details

Search Newyork taxi



NYC TLC Trips

[City of New York](#)

New York City taxi and limousine trips since 2009

[View dataset](#)

3

[Overview](#)

[Samples](#)

[Related Products](#)

Overview

This dataset is collected by the NYC Taxi and Limousine Commission (TLC) and includes trip records from all trips completed in Yellow and Green taxis in NYC, and all trips in for-hire vehicles (FHV) in the last 5 years. Records include fields capturing pick-up and drop-off dates/times, pick-up and drop-off locations, trip distances, itemized fares, rate types, payment types, and driver-reported passenger counts. For detailed information about this dataset, go to [TOC Trip Record Data](#)

This public dataset is hosted in Google BigQuery and is included in BigQuery's 1TB/mo of free tier processing. This means that each user receives 1TB of free BigQuery processing every month, which can be

Additional details

Type: [Data](#)

Category: [Encyclopedic](#)

Dataset source: [NYC OpenData](#)

Cloud service: BigQuery

Expected update frequency: Weekl



Sandbox Set up billing to upgrade to the full BigQuery experience. [Learn more](#)



Explorer

+ Add data

Search for resources

Home

Starred

Shared with me

Job history

▼ vaulted-art-485212-r1

Datasets

Connections

Conversations

Agents

Queries

(Classic) Queries

Notebooks

Data canvases

Data preparations

Pipelines

Untitled query

new_york...ips

bigquery-public-data / Datasets / new_york_taxi_trips

☆ new_york_taxi_trips

+ Create Tab

Overview

Details

Insights

Preview

tlc_green_trips_2021	Table	Sep 14, 2022, 11:11:50 AM UTC+7	None	None
tlc_green_trips_2022	Table	Sep 14, 2022, 11:11:54 AM UTC+7	None	None
tlc_green_trips_2023	Table	Mar 21, 2023, 7:06:18 AM UTC+7	None	None
tlc_yellow_trips_2011	Table	Sep 14, 2022, 9:22:04 PM UTC+7	None	None
tlc_yellow_trips_2012	Table	Sep 14, 2022, 9:19:08 PM UTC+7	None	None
tlc_yellow_trips_2013	Table	Sep 14, 2022, 11:16:39 AM UTC+7	None	None
tlc_yellow_trips_2014	Table	Sep 14, 2022, 11:16:06 AM UTC+7	None	None
tlc_yellow_trips_2015	Table	Sep 14, 2022, 11:15:52 AM UTC+7	None	None
tlc_yellow_trips_2016	Table	Sep 14, 2022, 11:15:29 AM UTC+7	None	None
tlc_yellow_trips_2017	Table	Sep 14, 2022, 11:15:10 AM UTC+7	None	None
tlc_yellow_trips_2018	Table	Sep 14, 2022, 11:14:54 AM UTC+7	None	None
tlc_yellow_trips_2019	Table	Sep 14, 2022, 11:14:34 AM UTC+7	None	None
tlc_yellow_trips_2020	Table	Sep 14, 2022, 11:14:16 AM UTC+7	None	None
tlc_yellow_trips_2021	Table	Sep 14, 2022, 11:13:56 AM UTC+7	None	None
tlc_yellow_trips_2022	Table	Sep 14, 2022, 11:13:35 AM UTC+7	None	None
tlc_yellow_trips_2023	Table	Mar 21, 2023, 7:09:09 AM UTC+7	None	None

4



Sandbox Set up billing to upgrade to the full BigQuery experience. [Learn more](#)



Explorer

[+ Add data](#)

Search for resources

Home

Starred

Shared with me

Job history

▼ vaulted-art-485212-r1

Datasets

Connections

Conversations

Agents

Queries

(Classic) Queries

Notebooks

Data canvases

Data preparations

Pipelines

Untitled query x **tlc_yello... 022** x +

bigquery-public-data / Datasets / new_york_taxi_trips / Tables / **tlc_yellow_trips_2022**

☆ **tlc_yellow_trips_2022**

[Query](#)

[Chat](#)

[Open in](#)

[+ Share](#)

[Copy](#)

[Schema](#)

[Details](#)

[Preview](#)

[Table Explorer](#)

Preview

[Insights](#)

[Lineage](#)

[Data Profile](#)

[Filter](#) Enter property name or value

☐	Field name	Type	Mode	Description	
☐	vendor_id	STRING	REQUIRED	A code indicating the LPEP pro...	▼
☐	pickup_datetime	TIMESTAMP	NULLABLE	The date and time when the me...	▼
☐	dropoff_datetime	TIMESTAMP	NULLABLE	The date and time when the me...	▼
☐	passenger_count	INTEGER	NULLABLE	The number of passengers in t...	▼
☐	trip_distance	NUMERIC	NULLABLE	The elapsed trip distance in mil...	▼
☐	rate_code	STRING	NULLABLE	The final rate code in effect at t...	▼
☐	store_and_fwd_flag	STRING	NULLABLE	This flag indicates whether the ...	▼
☐	payment_type	STRING	NULLABLE	A numeric code signifying how ...	▼
☐	fare_amount	NUMERIC	NULLABLE	The time-and-distance fare calc...	▼
☐	extra	NUMERIC	NULLABLE	Miscellaneous extras and surc...	▼
☐	mta_tax	NUMERIC	NULLABLE	\$0.50 MTA tax that is automati...	▼
☐	tip_amount	NUMERIC	NULLABLE	Tip amount. This field is autom...	▼

1 `SELECT * FROM `bigquery-public-data.new_york_taxi_trips.tlc_yellow_trips_2022` LIMIT 1000`

Query completed

Using on-demand processing quota

9:48 PM
SELECT * FROM `bi...

Job information				Results	Visualization	JSON	Execution details
Row	vendor_id	pickup_datetime	dropoff_datetime				
1	1	2022-02-10 13:07:20 UTC	2022-02-10 13:18:22 UTC				
2	1	2022-02-01 13:08:34 UTC	2022-02-01 13:08:37 UTC				
3	1	2022-02-03 14:47:23 UTC	2022-02-03 14:47:23 UTC				
4	2	2022-02-27 11:08:55 UTC	2022-02-27 11:09:40 UTC				
5	2	2022-02-28 14:42:44 UTC	2022-02-28 14:43:50 UTC				
6	1	2022-02-23 08:27:00 UTC	2022-02-23 08:27:00 UTC				

[tlc_yellow_trips_2022](#) Preview

bigquery-public-
data.new_york_taxi_trips.tlc_yellow_trips_2022
36.26M rows • 6.97 GB • Last modified Jan 26, 2023, 7:12 AM

Filter Enter property name or value

Field name	Type
vendor_id	STRING
pickup_datetime	TIMESTAMP
dropoff_datetime	TIMESTAMP
passenger_count	INTEGER
trip_distance	NUMERIC
rate_code	STRING
store_and_fwd_flag	STRING
payment_type	STRING
fare_amount	NUMERIC
extra	NUMERIC
mta_tax	NUMERIC
tip_amount	NUMERIC

Which is the most popular pickup_location_id?

```
SELECT
  pickup_location_id,
  COUNT(pickup_location_id) AS count_of_pickups
FROM
  `bigquery-public-data`.`new_york_taxi_trips`.`tlc_yellow_trips_2022`
GROUP BY
  pickup_location_id
ORDER BY
  COUNT(pickup_location_id) DESC
LIMIT
  1;
```


Add a new column that shows each trip duration in minutes.

```
SELECT
    pickup_datetime,
    dropoff_datetime,
    TIMESTAMP_DIFF(dropoff_datetime, pickup_datetime, SECOND) / 60 AS trip_duration_minutes
FROM
    `bigquery-public-data`.`new_york_taxi_trips`.`tlc_yellow_trips_2022`
ORDER BY
    pickup_datetime;
```

Q & A